Project Schedule Analysis

1. Task dependencies

Sprint 3 work spanned patient booking & payments, post-visit workflows (notes, ratings), therapist clinical views, admin operations, and a separate patient commerce item. Several items had clear sequencing; others could run in parallel across people (AZ, AY, JH, RL, YL).

Patient booking & pay (core journey)

- SCRUM-36 (Therapist List) enables patients to choose a provider before booking.
- SCRUM-39 (My Appointments) depends on appointment APIs and UI being available to list upcoming/past care. It builds on the same domain as booking, even if the initial booking UI was started earlier.
- SCRUM-38 (Payment + Confirmation) depends on a stable create/update appointment flow so "booking success" can be tied to payment state (paid vs pending) and confirmation UX.

Post-visit & therapist-facing

- SCRUM-30 (Session Notes after appointment) assumes appointments exist and typically follows core appointment lifecycle work (often after 39 is usable for therapists/patients).
- SCRUM-52 (Rate therapist after completed) depends on correct appointment status (e.g. Completed) and a surface to trigger rating—often downstream of 39 and lifecycle fixes.
- SCRUM-51 (Therapists view patient health metrics) depends on patient monitoring / health data being exposed via backend APIs and secured for therapist role—can proceed in parallel with payment once contracts are clear, but integration testing still couples it to shared auth/data patterns.

Admin chain

- SCRUM-47 (Admin content management dashboard) provides the admin shell / patterns for internal tools.
- SCRUM-48 (Admin bookings management) logically follows or tightly couples to 47 because it reuses admin navigation, auth, and table/list patterns.
- SCRUM-49 (Fix admin logout) is a cross-cutting fix; it can be done in parallel but should be verified against 47/48 flows so admins do not lose session state incorrectly.

Education chain

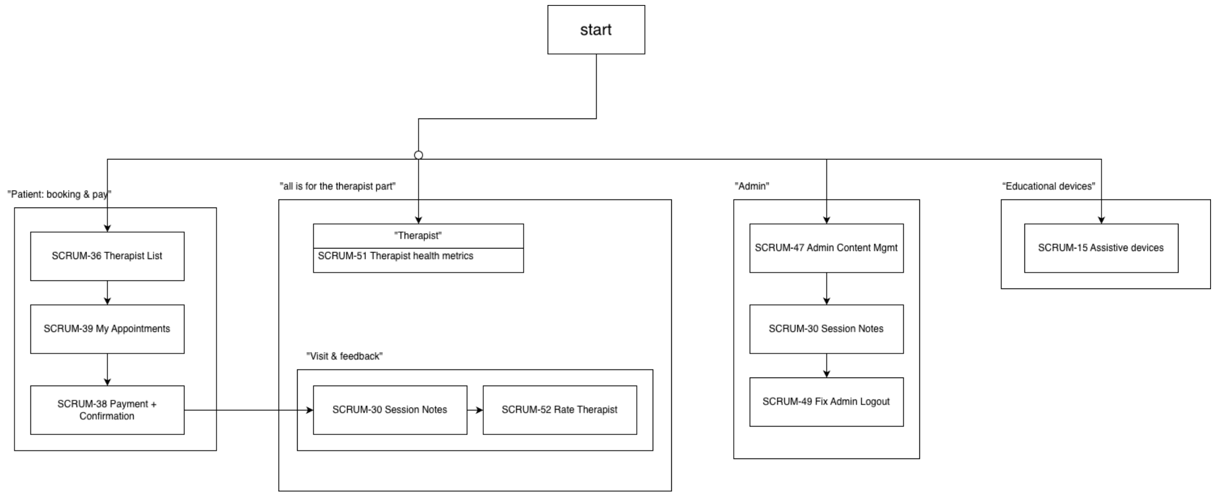
- SCRUM-15 (Purchase assistive devices) is largely separate from the appointment/payment critical path (different user journey), though it may reuse shared UI/auth.

Dependency chains (summary)

- Start → SCRUM-36 → SCRUM-39 → SCRUM-38
- Start → SCRUM-39 / SCRUM-30 (notes once appointments are live; partial overlap with 36→39 ordering in practice)
- Start → SCRUM-52 (after completed path & listing stable—often after 39 + lifecycle)
- Start → SCRUM-47 → SCRUM-48 ; SCRUM-49 in parallel with retest on admin pages
- Start → SCRUM-51 (parallel with patient chain after API/auth clarity)
- Start → SCRUM-15 (parallel track)

2. Network Diagram

The following simplified network diagram illustrates the logical sequence of development tasks during the sprint.



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3. Critical path

The critical path is the dependent sequence that most constrains demo-ready end-to-end value.

Primary (patient):

- SCRUM-36 → SCRUM-39 → SCRUM-38
- This is the longest single-user journey chain: pick therapist → see/track appointments → pay and confirm (38 is also the largest item at 5 SP). Delay here directly delays the core “care booking is real” milestone.

Secondary critical chains:

- SCRUM-47 → SCRUM-48 for admin operations (bookings oversight).
- SCRUM-52 depends on appointment completion semantics; it is critical for satisfaction/feedback but is shorter than the payment chain if pay is blocked.
- SCRUM-51 is critical for therapist trust but can run parallel to the patient pay path once APIs exist.

4. Risk areas

- Integration risk (38 + 39 + 36): payment touches API, redirects/mocks, and UI; any mismatch blocks “booking truly successful.”
- Data / permission risk (51, 52): therapist metrics and ratings require correct role checks and stable appointment states.
- Admin regression (47–49): logout fixes can break sessions for 48 if not retested together.
- Parallel work across 5 assignees: merge conflicts and API contract drift if branches stay separate too long.
- Sprint calendar: coursework peaks can compress integration to the end (consistent with burndown shape).
- Mitigations: early API contracts, smoke tests after merges, admin retest checklist for 49, and mid-sprint integration slot.

5. Lessons learned

- Drawing this dependency map during planning would have made 38’s reliance on 36/39 even clearer and reduced blocked time.
- 5-point stories (38) should be split earlier if payment has subtasks (initiate pay, pending, success page, mock vs real).
- Admin items benefit from one stable shell (47) before 48 expands surface area.
- SCRUM-15 staying separate in the diagram helps the team avoid accidentally coupling commerce to appointment deadlines.